

ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

CONTENTS

1. GATE CE 2026 (CE1)

A. Formulae

1.1.1 Intersection of Conics and Common Chord

- Conic in Vector Form: Every second-degree curve $ax^2 + 2bxy + cy^2 + 2dx + 2ey + f = 0$ can be written as

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0, \quad \mathbf{V} = \begin{pmatrix} a & b \\ b & c \end{pmatrix} = \mathbf{V}^\top, \quad \mathbf{u} = \begin{pmatrix} d \\ e \end{pmatrix} \quad (1.1.1.1)$$

The conic is a parabola if $|\mathbf{V}| = 0$, an ellipse if $|\mathbf{V}| > 0$ and a hyperbola if $|\mathbf{V}| < 0$. For a circle with centre $\mathbf{c}$ and radius r , $\|\mathbf{x} - \mathbf{c}\|^2 = r^2$ gives

$$\mathbf{V} = \mathbf{I}, \quad \mathbf{u} = -\mathbf{c}, \quad f = \|\mathbf{c}\|^2 - r^2 \quad (1.1.1.2)$$

- Line-Conic Intersection: The points of intersection of a line $L : \mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ ($\kappa \in \mathbb{R}$) with a conic section $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$ are given by

$$\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m} \quad (1.1.1.3)$$

where the parameter values κ_i are:

$$\kappa_i = \frac{-\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h}) (\mathbf{m}^\top \mathbf{V} \mathbf{m})}}{\mathbf{m}^\top \mathbf{V} \mathbf{m}} \quad (1.1.1.4)$$

with $g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V} \mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f$. The line meets the conic at two, one or no real points according as the discriminant

$$\Delta = [\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h}) (\mathbf{m}^\top \mathbf{V} \mathbf{m}) \quad (1.1.1.5)$$

is positive, zero or negative.

- Derivation of the Intersection Parameter: Substituting $\mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ into the conic and using $\mathbf{V} = \mathbf{V}^\top$,

$$(\mathbf{h} + \kappa \mathbf{m})^\top \mathbf{V} (\mathbf{h} + \kappa \mathbf{m}) + 2\mathbf{u}^\top (\mathbf{h} + \kappa \mathbf{m}) + f = 0 \quad (1.1.1.6)$$

$$\kappa^2 (\mathbf{m}^\top \mathbf{V} \mathbf{m}) + 2\kappa \mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) + g(\mathbf{h}) = 0 \quad (1.1.1.7)$$

This is a quadratic in κ , and the quadratic formula gives (??). Each real root κ_i gives one point of intersection through (??). When $\Delta = 0$ the two roots coincide and the line is a tangent to the conic.

- Pencil of Conics: The intersection of two conics with parameters $(\mathbf{V}_1, \mathbf{u}_1, f_1)$ and $(\mathbf{V}_2, \mathbf{u}_2, f_2)$ lies on every member of the pencil

$$\mathbf{x}^\top (\mathbf{V}_1 + \mu \mathbf{V}_2) \mathbf{x} + 2(\mathbf{u}_1 + \mu \mathbf{u}_2)^\top \mathbf{x} + (f_1 + \mu f_2) = 0 \quad (1.1.1.8)$$

- Degenerate Conic and Common Chord: The pencil (??) is a pair of straight lines (a degenerate conic) if and only if

$$\begin{vmatrix} \mathbf{V}_1 + \mu \mathbf{V}_2 & \mathbf{u}_1 + \mu \mathbf{u}_2 \\ (\mathbf{u}_1 + \mu \mathbf{u}_2)^\top & f_1 + \mu f_2 \end{vmatrix} = 0 \quad (1.1.1.9)$$

Every point common to both conics lies on every member of the pencil, so the common points lie on these lines. If $\mathbf{V}_1 = \mathbf{V}_2$ (for example, two circles), choosing $\mu = -1$ removes the second-degree terms and leaves the common chord (radical axis)

$$2(\mathbf{u}_1 - \mathbf{u}_2)^\top \mathbf{x} + (f_1 - f_2) = 0 \quad (1.1.1.10)$$

1.1.2 Line Intersection and Angle Between Vectors

- Forms of a Line: A line through the point $\mathbf{A}$ with direction vector $\mathbf{m}$ can be written in parametric form or, using a normal vector $\mathbf{n}$ with $\mathbf{n}^\top \mathbf{m} = 0$, in normal form:

$$\mathbf{x} = \mathbf{A} + \kappa \mathbf{m}, \quad \mathbf{n}^\top (\mathbf{x} - \mathbf{A}) = 0 \quad (1.1.2.1)$$

In two dimensions, if $\mathbf{m} = \begin{pmatrix} m_1 \\ m_2 \end{pmatrix}$ then $\mathbf{n} = \begin{pmatrix} -m_2 \\ m_1 \end{pmatrix}$.

- The lines $\mathbf{x} = \mathbf{A} + \kappa_1 \mathbf{m}_1$ and $\mathbf{x} = \mathbf{B} + \kappa_2 \mathbf{m}_2$ meet where

$$\begin{pmatrix} \mathbf{m}_1 & -\mathbf{m}_2 \end{pmatrix} \begin{pmatrix} \kappa_1 \\ \kappa_2 \end{pmatrix} = \mathbf{B} - \mathbf{A} \quad (1.1.2.2)$$

- Solvability: Equating the two parametric forms gives $\kappa_1 \mathbf{m}_1 - \kappa_2 \mathbf{m}_2 = \mathbf{B} - \mathbf{A}$, which is the matrix equation above. It has a unique solution if and only if $|\mathbf{m}_1 \quad -\mathbf{m}_2| \neq 0$, that is, the direction vectors are linearly independent. If the determinant is zero, the lines are parallel (no intersection) or coincident (infinitely many common points).
- The angle θ between two vectors $\mathbf{a}$ and $\mathbf{b}$ is given by

$$\cos \theta = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \quad (1.1.2.3)$$

- Dot Product Properties: The relation above follows from the law of cosines, $\|\mathbf{a} - \mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$, since $\|\mathbf{a} - \mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\mathbf{a}^\top \mathbf{b}$. Two vectors are orthogonal if and only if $\mathbf{a}^\top \mathbf{b} = 0$. The angle between two lines is the angle between their direction vectors.
- A point on a circle of radius r centred at the origin is $\mathbf{x} = r \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix}$, and

$$\mathbf{e}_\alpha^\top \mathbf{e}_\beta = \cos \alpha \cos \beta + \sin \alpha \sin \beta = \cos(\alpha - \beta), \quad \mathbf{e}_\alpha \triangleq \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix} \quad (1.1.2.4)$$

- Chords and Central Angles: For points $\mathbf{x}_\alpha = r \mathbf{e}_\alpha$ and $\mathbf{x}_\beta = r \mathbf{e}_\beta$ on the circle, the angle subtended at the centre is $\alpha - \beta$, and expanding the norm gives the chord length:

$$\|\mathbf{x}_\alpha - \mathbf{x}_\beta\|^2 = 2r^2 (1 - \mathbf{e}_\alpha^\top \mathbf{e}_\beta) = 4r^2 \sin^2 \frac{\alpha - \beta}{2} \quad (1.1.2.5)$$

The angle subtended by a chord at any point on the remaining arc is half the central angle.

1.1.3 System of Linear Equations and Rouché–Capelli Theorem

- A system of m linear equations in n variables is represented as

$$\mathbf{A} \mathbf{x} = \mathbf{b} \quad (1.1.3.1)$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{x} \in \mathbb{R}^{n \times 1}$ and $\mathbf{b} \in \mathbb{R}^{m \times 1}$. The system is consistent if and only if

$$\text{rank}(\mathbf{A}) = \text{rank}([\mathbf{A} \mid \mathbf{b}]) = r \quad (1.1.3.2)$$

- If $r = n$ the solution is unique; if $r < n$ there are infinitely many solutions with $n - r$ free variables. If $\mathbf{A}$ is square and non-singular, $\mathbf{x} = \mathbf{A}^{-1} \mathbf{b}$. If $\mathbf{A} \in \mathbb{R}^{n \times (n-1)}$, the system is consistent only if the square augmented matrix is singular:

$$|\mathbf{A} \quad \mathbf{b}| = 0 \quad (1.1.3.3)$$

- For $\mathbf{A} \in \mathbb{R}^{2 \times 3}$ with rank 2, the solution of $\mathbf{A} \mathbf{x} = \mathbf{0}$ is the line through the origin along the cross product of the two rows:

$$\mathbf{x} = \kappa (\mathbf{a}_1 \times \mathbf{a}_2) \quad (1.1.3.4)$$

- Zero Norm: For $\mathbf{v} \in \mathbb{R}^n$,

$$\|\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{v} = \sum_i v_i^2 = 0 \iff \mathbf{v} = \mathbf{0} \quad (1.1.3.5)$$

1.1.4 Relation Between Eigenvalues and Trace of a Matrix

- The characteristic polynomial of $\mathbf{A} \in \mathbb{R}^{n \times n}$ is

$$P(\lambda) = |\lambda \mathbf{I} - \mathbf{A}| \quad (1.1.4.1)$$

- The sum of all eigenvalues is equal to the trace:

$$\sum_{i=1}^n \lambda_i = \sum_{i=1}^n a_{ii} = \text{tr}(\mathbf{A}) \quad (1.1.4.2)$$

where $\mathbf{A}$ is symmetric if $\mathbf{A}^\top = \mathbf{A}$ and skew-symmetric if $\mathbf{A}^\top = -\mathbf{A}$.

1.1.5 Cholesky Decomposition

- A symmetric positive definite matrix $\mathbf{A}$ can be factorised as

$$\mathbf{A} = \mathbf{L}\mathbf{L}^\top \quad (1.1.5.1)$$

where $\mathbf{L}$ is lower triangular. For a 2×2 matrix,

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix} = \begin{pmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{pmatrix} \begin{pmatrix} l_{11} & l_{21} \\ 0 & l_{22} \end{pmatrix} = \begin{pmatrix} l_{11}^2 & l_{11}l_{21} \\ l_{11}l_{21} & l_{21}^2 + l_{22}^2 \end{pmatrix} \quad (1.1.5.2)$$

1.1.6 Bayes' Theorem in Vector Form

- Let $\boldsymbol{\pi} = (\text{Pr}(B_1) \ \cdots \ \text{Pr}(B_n))^\top$ be the prior vector and $\boldsymbol{\ell} = (\text{Pr}(E | B_1) \ \cdots \ \text{Pr}(E | B_n))^\top$ the likelihood vector. Then

$$\text{Pr}(E) = \boldsymbol{\pi}^\top \boldsymbol{\ell} \quad (1.1.6.1)$$

$$\text{Pr}(B_i | E) = \frac{\pi_i \ell_i}{\boldsymbol{\pi}^\top \boldsymbol{\ell}} \quad (1.1.6.2)$$

1.1.7 Minimization via Completing the Square

- For real a and b ,

$$ab - b^2 = \frac{a^2}{4} - \left(b - \frac{a}{2}\right)^2 \leq \frac{a^2}{4} \quad (1.1.7.1)$$

with equality if and only if $b = \frac{a}{2}$.

- If $g(x) \geq 0$ is continuous on $[a, b]$ and $\int_a^b g(x) dx = 0$, then $g(x) = 0$ on $[a, b]$.

1.1.8 Laplace Transform and State-Space Form

- From the derivative property,

$$y'(t) \xleftrightarrow{\mathcal{L}} sY(s) - y(0) \quad (1.1.8.1)$$

$$y''(t) \xleftrightarrow{\mathcal{L}} s^2Y(s) - sy(0) - y'(0) \quad (1.1.8.2)$$

$$e^{at}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s-a} \quad (1.1.8.3)$$

- Writing $\mathbf{z} = (y \ y')^\top$, the equation $y'' + py' + qy = 0$ becomes

$$\mathbf{z}' = \mathbf{A}\mathbf{z}, \quad \mathbf{A} = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix} \quad (1.1.8.4)$$

and the exponents of the solution are the eigenvalues of $\mathbf{A}$.

1.1.9 Newton-Raphson Method

- For a differentiable $f(x)$, the root is approximated iteratively by

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (1.1.9.1)$$

1.1.10 Polynomial Interpolation (Vandermonde Matrix)

- The polynomial $P_2(x) = c_0 + c_1x + c_2x^2$ through (x_i, y_i) , $i = 1, 2, 3$, satisfies

$$\begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \quad (1.1.10.1)$$

1.1.11 Equilibrium of Rigid Bodies in Vector Form

- For a body in equilibrium under forces $\mathbf{F}_i$ acting at points $\mathbf{r}_i$,

$$\sum_i \mathbf{F}_i = \mathbf{0}, \quad \sum_i (\mathbf{r}_i - \mathbf{O}) \times \mathbf{F}_i = \mathbf{0} \quad (1.1.11.1)$$

where in the plane $\mathbf{r} \times \mathbf{F} = r_x F_y - r_y F_x$.

- The bending moment at a section $\mathbf{P}$ of a member is the moment about $\mathbf{P}$ of all forces on one side of the section:

$$M(\mathbf{P}) = \sum_{\text{one side}} (\mathbf{r}_i - \mathbf{P}) \times \mathbf{F}_i \quad (1.1.11.2)$$

- When sliding impends, the limiting friction force is $F = \mu N$.

1.1.12 Direct Stiffness Method

- For an axial member of length L , area A , modulus E , with unit direction vector $\mathbf{m}$, the stiffness matrix of its end node is

$$\mathbf{k} = \frac{AE}{L} \mathbf{m} \mathbf{m}^\top \quad (1.1.12.1)$$

and the member stiffness matrix is $\begin{pmatrix} \mathbf{k} & -\mathbf{k} \\ -\mathbf{k} & \mathbf{k} \end{pmatrix}$.

- The global stiffness matrix is assembled by adding member matrices at the corresponding degrees of freedom:

$$\mathbf{K} \mathbf{u} = \mathbf{F} \quad (1.1.12.2)$$

- Boundary conditions are applied by deleting the rows and columns of restrained degrees of freedom. A temperature rise ΔT in a member with coefficient α is equivalent to nodal loads of magnitude $AE\alpha\Delta T$ at its ends, directed outward.

1.1.13 Cellular Update Rule in Matrix Form

- For N bulbs in a line with state vector $\mathbf{x}_k \in \{0, 1\}^N$ ($1 = \text{ON}$), the number of ON neighbours is

$$\mathbf{s}_k = \mathbf{A} \mathbf{x}_k, \quad a_{ij} = \begin{cases} 1, & |i - j| = 1 \\ 0, & \text{otherwise} \end{cases} \quad (1.1.13.1)$$

1.1.14 Surveying and Traffic Relations

- Reduced level: With height of instrument (plane of collimation) HI and staff reading h ,

$$RL = HI - h \quad (1.1.14.1)$$

- Space mean speed: For n vehicles covering distance d in times $\mathbf{t} = (t_1 \ \cdots \ t_n)^\top$,

$$v_s = \frac{nd}{\mathbf{1}^\top \mathbf{t}} = \frac{n}{\sum_i \frac{1}{v_i}} \quad (1.1.14.2)$$

which is the harmonic mean of the spot speeds $v_i = d/t_i$.

B. Problems

1.2.1 At how many points will the curves $y = x^2$ and $y = -x^2 - 2x - 1$ intersect in the real (x, y) plane? (GATE CE 2026)

- | | |
|------|------|
| a) 0 | c) 2 |
| b) 1 | d) 3 |

Solution: The given curves can be written in the form $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$:

$$x^2 - y = 0 \implies \mathbf{V}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{u}_1 = \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}, \quad f_1 = 0 \quad (1.2.1.1)$$

$$x^2 + 2x + y + 1 = 0 \implies \mathbf{V}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix}, \quad f_2 = 1 \quad (1.2.1.2)$$

Choosing $\mu = -1$ in the pencil (??) eliminates the second-degree terms ($\mathbf{V}_1 - \mathbf{V}_2 = \mathbf{0}$):

$$2(\mathbf{u}_1 - \mathbf{u}_2)^\top \mathbf{x} + (f_1 - f_2) = 0 \quad (1.2.1.3)$$

$$2(-1 \ -1) \mathbf{x} - 1 = 0 \implies (1 \ 1) \mathbf{x} = -\frac{1}{2} \quad (1.2.1.4)$$

All common points of the two curves lie on this line. A point on the line and its direction vector are

$$\mathbf{h} = \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (1.2.1.5)$$

Substituting into (??) for the first curve:

$$\mathbf{m}^\top \mathbf{V}_1 \mathbf{m} = (1 \ -1) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1 \quad (1.2.1.6)$$

$$\mathbf{V}_1 \mathbf{h} + \mathbf{u}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix} \quad (1.2.1.7)$$

$$\mathbf{m}^\top (\mathbf{V}_1 \mathbf{h} + \mathbf{u}_1) = (1 \ -1) \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix} = \frac{1}{2} \quad (1.2.1.8)$$

$$g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V}_1 \mathbf{h} + 2\mathbf{u}_1^\top \mathbf{h} + f_1 = 0 + \frac{1}{2} + 0 = \frac{1}{2} \quad (1.2.1.9)$$

$$\Delta = \left(\frac{1}{2}\right)^2 - \frac{1}{2}(1) = -\frac{1}{4} < 0 \quad (1.2.1.10)$$

Since the discriminant is negative, κ in (??) is not real and the curves do not meet in the real plane.

codes/ce1/q03.py
figs/q03.pdf

a) 40°
b) 50°

c) 60°
d) 80°

$$\mathbf{P} = r\mathbf{e}_{180^\circ} = -r\mathbf{e}_0, \quad \mathbf{Q} = r\mathbf{e}_0, \quad \mathbf{R} = r\mathbf{e}_\alpha, \quad \mathbf{S} = r\mathbf{e}_\beta \quad (1.2.2.1)$$
$$\cos \angle ROS = \mathbf{e}_\alpha^\top \mathbf{e}_\beta = \cos(\alpha - \beta) \implies \alpha - \beta = 80^\circ \quad (1.2.2.2)$$
$$\mathbf{R} - \mathbf{P} = r \begin{pmatrix} 1 + \cos \alpha \\ \sin \alpha \end{pmatrix} = 2r \cos \frac{\alpha}{2} \begin{pmatrix} \cos \frac{\alpha}{2} \\ \sin \frac{\alpha}{2} \end{pmatrix} \quad (1.2.2.3)$$

$$\mathbf{S} - \mathbf{Q} = r \begin{pmatrix} \cos \beta - 1 \\ \sin \beta \end{pmatrix} = 2r \sin \frac{\beta}{2} \begin{pmatrix} -\sin \frac{\beta}{2} \\ \cos \frac{\beta}{2} \end{pmatrix} \quad (1.2.2.4)$$

$$\cos \angle RTS = \left(\cos \frac{\alpha}{2} \quad \sin \frac{\alpha}{2} \right) \begin{pmatrix} -\sin \frac{\beta}{2} \\ \cos \frac{\beta}{2} \end{pmatrix} \quad (1.2.25)$$

$$= \sin \frac{\alpha}{2} \cos \frac{\beta}{2} - \cos \frac{\alpha}{2} \sin \frac{\beta}{2} = \sin \frac{\alpha - \beta}{2} \quad (1.2.2.6)$$

$$= \sin 40^\circ = \cos 50^\circ \quad (1.2.2.7)$$

$$\implies \angle RTS = 50^\circ \quad (1.2.2.8)$$

For a numerical check, take $r = 1$, $\alpha = 130^\circ$, $\beta = 50^\circ$. From (??),

$$(\mathbf{R} - \mathbf{P} - (\mathbf{S} - \mathbf{Q})) \begin{pmatrix} \kappa_1 \\ \kappa_2 \end{pmatrix} = \mathbf{Q} - \mathbf{P} \implies \mathbf{T} = \begin{pmatrix} 0 \\ 2.1445 \end{pmatrix} \quad (1.2.2.9)$$

and (??) gives $\angle RTS = 50^\circ$.

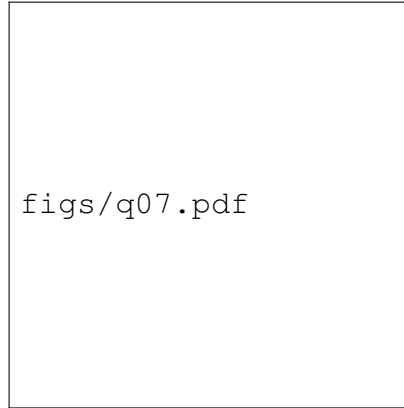


Fig. 1.2.2.1. Circle with diameter PQ , $\angle ROS = 80^\circ$ and $\angle RTS = 50^\circ$

codes/ce1/q07.py

figs/q07.pdf

- 1.2.3 Consider a linear arrangement of seven bulbs, each of which can be in the ON or OFF states. The initial configuration of the bulbs is shown in the figure. In every Step, the states of the bulbs are changed based on the following rules: Any OFF bulb with exactly one ON neighbor at the end of the previous Step is turned ON. Any ON bulb with both neighbors ON at the end of the previous Step is turned OFF. The state of any bulb not meeting the conditions above is left unchanged. The state of bulbs at the end of Step 1 and Step 2 are also shown in the figure. The number of bulbs which are ON at the end of Step 8 is _____ (GATE CE 2026)

- a) 5
b) 4
c) 3
d) 0

Solution: Let $\mathbf{x}_k \in \{0, 1\}^7$ be the state vector after Step k and $\mathbf{s}_k = \mathbf{A}\mathbf{x}_k$ from (??), where

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \quad (1.2.3.1)$$

The rules give the update

$$\mathbf{x}_{k+1} = \mathbf{x}_k + (\mathbf{1} - \mathbf{x}_k) \odot [\mathbf{s}_k = 1] - \mathbf{x}_k \odot [\mathbf{s}_k = 2] \quad (1.2.3.2)$$

where $\odot$ is the element-wise product and $[\cdot]$ is 1 when the condition holds. The initial configuration has only the middle bulb ON:

$$\mathbf{x}_0 = (0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)^\top, \quad \mathbf{s}_0 = (0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0)^\top \quad (1.2.3.3)$$

$$\mathbf{x}_1 = (0 \ 0 \ 1 \ 1 \ 1 \ 0 \ 0)^\top, \quad \mathbf{s}_1 = (0 \ 1 \ 1 \ 2 \ 1 \ 1 \ 0)^\top \quad (1.2.3.4)$$

$$\mathbf{x}_2 = (0 \ 1 \ 1 \ 0 \ 1 \ 1 \ 0)^\top, \quad \mathbf{s}_2 = (1 \ 1 \ 1 \ 2 \ 1 \ 1 \ 1)^\top \quad (1.2.3.5)$$

$$\mathbf{x}_3 = (1 \ 1 \ 1 \ 0 \ 1 \ 1 \ 1)^\top, \quad \mathbf{s}_3 = (1 \ 2 \ 1 \ 2 \ 1 \ 2 \ 1)^\top \quad (1.2.3.6)$$

$$\mathbf{x}_4 = (1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 1)^\top, \quad \mathbf{s}_4 = (0 \ 2 \ 0 \ 2 \ 0 \ 2 \ 0)^\top \quad (1.2.3.7)$$

$\mathbf{x}_1$ and $\mathbf{x}_2$ agree with Step 1 and Step 2 in the figure. In $\mathbf{x}_4$, every OFF bulb has two ON neighbours and every ON bulb has none, so (??) gives

$$\mathbf{x}_5 = \mathbf{x}_4 \implies \mathbf{x}_8 = \mathbf{x}_4 = (1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 1)^\top \quad (1.2.3.8)$$

The number of ON bulbs after Step 8 is

$$\mathbf{1}^\top \mathbf{x}_8 = 4 \quad (1.2.3.9)$$

figs/q09.pdf

Fig. 1.2.3.1. Bulb states from the initial configuration to Step 8

codes/ce1/q09.py

figs/q09.pdf

1.2.4 P and Q are two positive integers such that $P^2 = Q^2 + 13$. The product of the numbers P and Q is _____ (GATE CE 2026)

a) 13

c) 39

b) 26

d) 42

Solution: The given equation can be written as

$$P^2 - Q^2 = (P - Q)(P + Q) = 13 \quad (1.2.4.1)$$

Since 13 is prime and $0 < P - Q < P + Q$,

$$P - Q = 1, \quad P + Q = 13 \quad (1.2.4.2)$$

Expressing in the matrix form (??):

$$\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} P \\ Q \end{pmatrix} = \begin{pmatrix} 1 \\ 13 \end{pmatrix} \quad (1.2.4.3)$$

Applying $R_2 \rightarrow R_2 - R_1$ on the augmented matrix:

$$\begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 13 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{pmatrix} 1 & -1 & 1 \\ 0 & 2 & 12 \end{pmatrix} \Rightarrow Q = 6, P = 7 \quad (1.2.4.4)$$

Therefore,

$$PQ = 7 \times 6 = 42 \quad (1.2.4.5)$$

codes/ce1/q10.py

1.2.5 Matrix $\mathbf{P}$ is given as

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \quad (1.2.5.1)$$

The TRUE option is

(GATE CE 2026)

- a) Trace of $\mathbf{P}$ is equal to the sum of the Eigen values of $\mathbf{P}$. c) $\mathbf{P}$ is a skew-symmetric matrix.
b) $\mathbf{P}^T \mathbf{P}$ is an identity matrix. d) Absolute magnitude of each Eigen value is 1.

Solution: From (??), expanding along the second row,

$$|\lambda \mathbf{I} - \mathbf{P}| = \begin{vmatrix} \lambda - 1 & 0 & -1 \\ 0 & \lambda - 1 & 0 \\ -1 & 0 & \lambda - 1 \end{vmatrix} \quad (1.2.5.2)$$

$$= (\lambda - 1) [(\lambda - 1)^2 - 1] = \lambda (\lambda - 1) (\lambda - 2) = 0 \quad (1.2.5.3)$$

$$\Rightarrow \lambda_1 = 0, \lambda_2 = 1, \lambda_3 = 2 \quad (1.2.5.4)$$

From (??):

$$\text{tr}(\mathbf{P}) = 1 + 1 + 1 = 3 = \lambda_1 + \lambda_2 + \lambda_3 \quad (1.2.5.5)$$

so option (A) is true. The remaining options are false since

$$\mathbf{P}^T \mathbf{P} = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 1 & 0 \\ 2 & 0 & 2 \end{pmatrix} \neq \mathbf{I}, \quad \mathbf{P}^T = \mathbf{P} \neq -\mathbf{P}, \quad |\lambda_1| = 0 \neq 1 \quad (1.2.5.6)$$

codes/ce1/q11.py

1.2.6 Given:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (1.2.6.1)$$

The above system of equations represents a

(GATE CE 2026)

- a) plane
b) line

- c) volume
d) point

Solution: The augmented matrix is reduced as

$$\begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 2 & 0 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & -1 & 1 & 0 \end{pmatrix} \quad (1.2.6.2)$$

Thus,

$$\text{rank}(\mathbf{A}) = \text{rank}([\mathbf{A} \mid \mathbf{b}]) = 2 < n = 3 \quad (1.2.6.3)$$

From (??), the system is consistent with $n - r = 1$ free variable. From (??), the solution is

$$\mathbf{x} = \kappa \left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \right) = \kappa \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}, \quad \kappa \in \mathbb{R} \quad (1.2.6.4)$$

which is a line through the origin (the intersection of two planes).



Fig. 1.2.6.1. The two planes and their line of intersection

codes/ce1/q12.py

figs/q12.pdf

- 1.2.7 Bag I contains 4 white and 6 black balls. Bag II contains 4 white and 3 black balls. One ball is drawn at random from any one of the two bags and it is found to be a black ball. The probability that the black ball was drawn from Bag I is _____ (rounded off to two decimal places). (GATE CE 2026)

Solution: Let B_1 , B_2 denote choosing Bag I, Bag II and E the event of drawing a black ball. The prior and likelihood vectors are

$$\boldsymbol{\pi} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}, \quad \boldsymbol{\ell} = \begin{pmatrix} \frac{6}{10} \\ \frac{3}{7} \end{pmatrix} \quad (1.2.7.1)$$

From (??),

$$\Pr(E) = \boldsymbol{\pi}^\top \boldsymbol{\ell} = \frac{1}{2} \left(\frac{3}{5} + \frac{3}{7} \right) = \frac{18}{35} \quad (1.2.7.2)$$

From (??),

$$\Pr(B_1 | E) = \frac{\frac{1}{2} \times \frac{3}{5}}{\frac{18}{35}} = \frac{3}{10} \times \frac{35}{18} = \frac{7}{12} \approx 0.58 \quad (1.2.7.3)$$

codes/ce1/q26.py

1.2.8 A matrix is given as:

$$\begin{pmatrix} 9 & 15 \\ 15 & 50 \end{pmatrix} \quad (1.2.8.1)$$

By performing Cholesky decomposition, $|l_{22}|$ of the lower triangular matrix is _____ (in integer). (GATE CE 2026)

Solution: Comparing with (??),

$$l_{11}^2 = 9 \implies l_{11} = 3 \quad (1.2.8.2)$$

$$l_{11}l_{21} = 15 \implies l_{21} = 5 \quad (1.2.8.3)$$

$$l_{21}^2 + l_{22}^2 = 50 \implies l_{22}^2 = 25 \implies |l_{22}| = 5 \quad (1.2.8.4)$$

Thus,

$$\mathbf{L} = \begin{pmatrix} 3 & 0 \\ 5 & 5 \end{pmatrix}, \quad \mathbf{L}\mathbf{L}^\top = \begin{pmatrix} 9 & 15 \\ 15 & 50 \end{pmatrix} \quad (1.2.8.5)$$

codes/ce1/q27.py

1.2.9 It is given that x and y are integers in the following equation:

$$(x + y - 7)^2 + (y + 3x - 13)^2 = 0 \quad (1.2.9.1)$$

The value of $(x^3 + y^3)$ is _____ (in integer). (GATE CE 2026)

Solution: Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 7 \\ 13 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \quad (1.2.9.2)$$

The given equation is

$$\|\mathbf{Ax} - \mathbf{b}\|^2 = 0 \quad (1.2.9.3)$$

From (??),

$$\mathbf{Ax} = \mathbf{b} \quad (1.2.9.4)$$

Reducing the augmented matrix:

$$\begin{pmatrix} 1 & 1 & 7 \\ 3 & 1 & 13 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \begin{pmatrix} 1 & 1 & 7 \\ 0 & -2 & -8 \end{pmatrix} \implies y = 4, x = 3 \quad (1.2.9.5)$$

Therefore,

$$x^3 + y^3 = 27 + 64 = 91 \quad (1.2.9.6)$$

codes/ce1/q28.py

Equating with the given value,

$$\int_0^2 \left(f(x) - \frac{x}{2} \right)^2 dx = 0 \quad (1.2.12.5)$$

Since the integrand is continuous and non-negative,

$$f(x) = \frac{x}{2} \implies f(1) = \frac{1}{2} \quad (1.2.12.6)$$

figs/q36.pdf

Fig. 1.2.12.1. The integrand attains its upper bound $\frac{x^2}{4}$ only for $f(x) = \frac{x}{2}$

codes/ce1/q36.py

figs/q36.pdf

1.2.13 An ordinary differential equation is given below.

$$x^2 \frac{d^2 y}{dx^2} = 6y \quad (1.2.13.1)$$

Considering a and b as arbitrary constants, the general solution of the equation is (GATE CE 2026)

- a) $y(x) = ax^3 + \frac{b}{x^2}$ c) $y(x) = ax^2 + b \ln x$
b) $y(x) = ax^2 + \frac{b}{x^3}$ d) $y(x) = ax^3 + b \ln x$

Solution: Let $x = e^t$ ($x > 0$). Then

$$x \frac{dy}{dx} = \frac{dy}{dt}, \quad x^2 \frac{d^2 y}{dx^2} = \frac{d^2 y}{dt^2} - \frac{dy}{dt} \quad (1.2.13.2)$$

and the equation becomes

$$y'' - y' - 6y = 0 \quad (1.2.13.3)$$

Taking the Laplace transform using (??) and (??):

$$s^2 Y(s) - sy(0) - y'(0) - [sY(s) - y(0)] - 6Y(s) = 0 \quad (1.2.13.4)$$

$$Y(s) = \frac{(s-1)y(0) + y'(0)}{s^2 - s - 6} = \frac{(s-1)y(0) + y'(0)}{(s-3)(s+2)} \quad (1.2.13.5)$$

Decomposing into partial fractions and using (??),

$$Y(s) = \frac{a}{s-3} + \frac{b}{s+2} \implies y(t) = ae^{3t} + be^{-2t} \quad (1.2.13.6)$$

Equivalently, from (??), the exponents are the eigenvalues of

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 6 & 1 \end{pmatrix}, \quad |\lambda \mathbf{I} - \mathbf{A}| = \lambda^2 - \lambda - 6 = 0 \implies \lambda = 3, -2 \quad (1.2.13.7)$$

Substituting $e^t = x$:

$$y(x) = ax^3 + \frac{b}{x^2} \quad (1.2.13.8)$$

codes/ce1/q37.py

- 1.2.14 A plane truss consists of two linearly elastic, homogeneous, identical members, namely PQ and QR . Both members have length (L), cross-sectional area (A), and modulus of elasticity (E). The members are inclined at 45° as shown in the figure. The truss has hinge supports at P and R . The translational degrees-of-freedom (u and v) are shown at joint Q .

After application of the boundary conditions, the stiffness matrix of the truss becomes: (GATE CE 2026)

- a) $\frac{AE}{L} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ c) $\frac{AE}{L} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
b) $\frac{AE}{L} \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$ d) $\frac{AE}{L} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$

Solution: The unit direction vectors of the members are

$$\mathbf{m}_{PQ} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{m}_{QR} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (1.2.14.1)$$

From (??),

$$\mathbf{k}_{PQ} = \frac{AE}{L} \mathbf{m}_{PQ} \mathbf{m}_{PQ}^\top = \frac{AE}{L} \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix} \quad (1.2.14.2)$$

$$\mathbf{k}_{QR} = \frac{AE}{L} \mathbf{m}_{QR} \mathbf{m}_{QR}^\top = \frac{AE}{L} \begin{pmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{pmatrix} \quad (1.2.14.3)$$

After deleting the restrained degrees of freedom at P and R , only the (u, v) block at Q remains, which receives contributions from both members:

$$\mathbf{K} = \mathbf{k}_{PQ} + \mathbf{k}_{QR} = \frac{AE}{L} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (1.2.14.4)$$

In column BD , the roller reaction is vertical and along the column, so $M = 0$. The maximum bending moment is 165 kN m under the 90 kN load.

figs/q39.pdf

Fig. 1.2.15.1. Frame loading and bending moment diagram

codes/ce1/q39.py

figs/q39.pdf

- 1.2.16 Starting with the first approximation as $x = 0.5$, the second approximation for the root of the following function by the Newton-Raphson method is _____ (rounded off to two decimal places).

$$f(x) = e^{-x} - x \quad (1.2.16.1)$$

(GATE CE 2026)

Solution: The function and its derivative are

$$f(x) = e^{-x} - x \quad (1.2.16.2)$$

$$f'(x) = -e^{-x} - 1 \quad (1.2.16.3)$$

Substituting into (??):

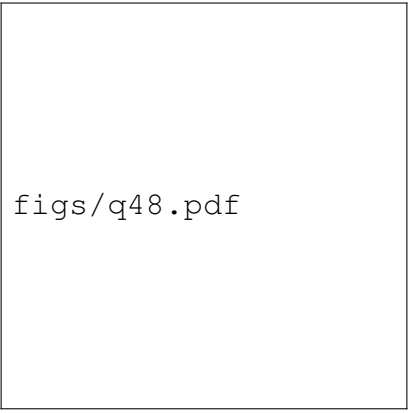
$$x_{n+1} = x_n + \frac{e^{-x_n} - x_n}{e^{-x_n} + 1} \quad (1.2.16.4)$$

$$= \frac{1 + x_n}{1 + e^{x_n}} \quad (1.2.16.5)$$

With the first approximation $x_0 = 0.5$, the second approximation is

$$x_1 = \frac{1.5}{1 + e^{0.5}} = \frac{1.5}{2.64872} = 0.56631 \approx 0.57 \quad (1.2.16.6)$$

Subsequent iterations converge to $x_2 = x_3 = 0.567143$.



figs/q48.pdf

Fig. 1.2.16.1. One Newton-Raphson step for $f(x) = e^{-x} - x$ from $x_0 = 0.5$

codes/ce1/q48.py

figs/q48.pdf

1.2.17 Values of y for different values of x are tabulated below.

x	-2	1	2
y	28	4	16

If a second-degree interpolating polynomial $P_2(x)$ is used to represent y , the value of $P_2(0)$ is _____ (rounded off to the nearest integer). (GATE CE 2026)

Solution: From (??),

$$\begin{pmatrix} 1 & -2 & 4 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 28 \\ 4 \\ 16 \end{pmatrix} \quad (1.2.17.1)$$

Reducing the augmented matrix:

$$\begin{pmatrix} 1 & -2 & 4 & 28 \\ 1 & 1 & 1 & 4 \\ 1 & 2 & 4 & 16 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 - R_1} \begin{pmatrix} 1 & -2 & 4 & 28 \\ 0 & 3 & -3 & -24 \\ 0 & 4 & 0 & -12 \end{pmatrix} \quad (1.2.17.2)$$

$$\implies c_1 = -3, c_2 = 5, c_0 = 2 \quad (1.2.17.3)$$

Thus,

$$P_2(x) = 5x^2 - 3x + 2 \implies P_2(0) = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \end{pmatrix} = 2 \quad (1.2.17.4)$$

figs/q49.pdf

Fig. 1.2.17.1. Interpolating polynomial through the given points

codes/ce1/q49.py

figs/q49.pdf

- 1.2.18 Two identical blocks A and B are connected by a rigid rod. The blocks rest against vertical and horizontal planes, as shown in the figure. The coefficient of static friction at the vertical and horizontal planes is the same. If the sliding impends when $\theta = 45^\circ$, the value of the coefficient of static friction is _____ (rounded off to two decimal places). (GATE CE 2026)

Solution: Let each block have weight W and let the rod make angle θ with the horizontal, with $c = \cos \theta$, $s = \sin \theta$. The rod is a two-force member carrying compression C along

$$\mathbf{e} = \frac{\mathbf{A} - \mathbf{B}}{\|\mathbf{A} - \mathbf{B}\|} = \begin{pmatrix} -c \\ s \end{pmatrix} \quad (1.2.18.1)$$

When sliding impends, A moves down and B moves away from the wall, so the friction forces are μN_A (upward on A) and μN_B (towards the wall on B). From (??):

$$\text{Block } A : \begin{pmatrix} N_A \\ \mu N_A \end{pmatrix} + \begin{pmatrix} 0 \\ -W \end{pmatrix} + C \begin{pmatrix} -c \\ s \end{pmatrix} = \mathbf{0} \quad (1.2.18.2)$$

$$\text{Block } B : \begin{pmatrix} -\mu N_B \\ N_B \end{pmatrix} + \begin{pmatrix} 0 \\ -W \end{pmatrix} - C \begin{pmatrix} -c \\ s \end{pmatrix} = \mathbf{0} \quad (1.2.18.3)$$

This gives four equations in three unknowns $(N_A \ N_B \ C)^\top$:

$$\begin{pmatrix} 1 & 0 & -c \\ \mu & 0 & s \\ 0 & -\mu & c \\ 0 & 1 & -s \end{pmatrix} \begin{pmatrix} N_A \\ N_B \\ C \end{pmatrix} = \begin{pmatrix} 0 \\ W \\ 0 \\ W \end{pmatrix} \quad (1.2.18.4)$$

From (??), the system is consistent only if

$$\begin{vmatrix} 1 & 0 & -c & 0 \\ \mu & 0 & s & W \\ 0 & -\mu & c & 0 \\ 0 & 1 & -s & W \end{vmatrix} = 0 \quad (1.2.18.5)$$

Expanding along the second column,

$$\mu \begin{vmatrix} 1 & -c & 0 \\ \mu & s & W \\ 0 & -s & W \end{vmatrix} + \begin{vmatrix} 1 & -c & 0 \\ \mu & s & W \\ 0 & c & 0 \end{vmatrix} = 0 \quad (1.2.18.6)$$

$$\mu W (2s + \mu c) - Wc = 0 \quad (1.2.18.7)$$

$$\Rightarrow \tan \theta = \frac{1 - \mu^2}{2\mu} \quad (1.2.18.8)$$

For $\theta = 45^\circ$:

$$\mu^2 + 2\mu - 1 = 0 \Rightarrow \mu = -1 + \sqrt{2} = 0.414 \quad (1.2.18.9)$$

taking the positive root.

figs/q50.pdf

Fig. 1.2.18.1. Blocks A and B connected by a rigid rod at $\theta = 45^\circ$

codes/ce1/q50.py

figs/q50.pdf

- 1.2.19 A homogenous, linearly elastic rod AB is connected to a linearly elastic spring BC in between the fixed supports at A and C , as shown in the figure. The cross-sectional area, modulus of elasticity, and the coefficient of thermal expansion of the rod AB are 500 mm^2 , $60 \times 10^3 \text{ MPa}$, and 12×10^{-6} per $^\circ\text{C}$, respectively. The stiffness (k) of spring BC is 2500 N/mm . The internal force (in kN) that will develop in the spring BC when the temperature of rod AB is increased by 100°C is _____ (rounded off to one decimal place). (GATE CE 2026)

Solution: From the figure, the length of rod AB is $L = 3 \text{ m} = 3000 \text{ mm}$. The rod stiffness is

$$k_r = \frac{AE}{L} = \frac{500 \times 60 \times 10^3}{3000} = 10^4 \text{ N/mm} \quad (1.2.19.1)$$

The equivalent thermal load is

$$F_{th} = AE\alpha\Delta T = 500 \times 60 \times 10^3 \times 12 \times 10^{-6} \times 100 = 36000 \text{ N} \quad (1.2.19.2)$$

Assembling (??) for nodes A, B, C :

$$\begin{pmatrix} k_r & -k_r & 0 \\ -k_r & k_r + k & -k \\ 0 & -k & k \end{pmatrix} \begin{pmatrix} u_A \\ u_B \\ u_C \end{pmatrix} = \begin{pmatrix} -F_{th} + R_A \\ F_{th} \\ R_C \end{pmatrix} \quad (1.2.19.3)$$

With $u_A = u_C = 0$, deleting the first and third rows and columns:

$$(k_r + k) u_B = F_{th} \implies u_B = \frac{36000}{10^4 + 2500} = 2.88 \text{ mm} \quad (1.2.19.4)$$

The force in the spring is

$$F_{BC} = k u_B = 2500 \times 2.88 = 7200 \text{ N} = 7.2 \text{ kN} \quad (1.2.19.5)$$



figs/q51.pdf

Fig. 1.2.19.1. Force in spring BC versus temperature rise of rod AB

/ce1/q51.py

figs/q51.pdf
